

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

Code of Conduct:

*I **Trisha N 192311398** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Trisha N

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding

Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 1: Responsive Web Design

Responsive Web Design (RWD) is an approach that allows a website to automatically adapt to different screen sizes such as mobile phones, tablets, laptops, and desktop computers. It provides an optimal viewing experience without horizontal scrolling or zooming.

Importance:

- Better user experience.
- Mobile-friendly interface.
- Improved SEO.
- Single website for all devices.
- Easier maintenance.

Key CSS techniques:

- Fluid layouts using percentages.
- Flexible images (max-width:100%).
- Media queries.
- Relative units (% , rem, em, vw, vh).
- Viewport meta tag.

Flexbox example:

HTML:

```
<div class="container">
  <div class="box">Home</div>
  <div class="box">About</div>
  <div class="box">Services</div>
```

```
<div class="box">Contact</div>
</div>
```

CSS:

```
.container{display:flex;flex-wrap:wrap;}
.box{flex:1;padding:20px;}
@media(max-width:768px){.container{flex-direction:column;}}
```

Improvements:

- Optimize images.
- Touch-friendly buttons.
- Responsive navigation.
- Readable fonts.
- Test on multiple devices.

Conclusion:

Responsive Web Design ensures websites are attractive, accessible, and functional on all devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security

Question 2: Client-Side Validation

Client-side validation checks user input in the browser before sending it to the server. It is implemented using JavaScript and HTML5 validation.

Purpose:

- Prevent invalid data.
- Provide instant feedback.
- Reduce server load.
- Improve usability.

JavaScript event handling:

Common events include onsubmit, oninput, onchange, onblur, and onclick.

Example:

```
document.getElementById("form").onsubmit=function(){
  let email=document.getElementById("email").value;
  let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  if(!pattern.test(email)){
    alert("Invalid Email");
    return false;
  }
}
```

Regular Expressions:

Email: /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[A-Za-z]{2,}\$/

Phone: /^[0-9]{10}\$/

Improvements:

- Show field-level error messages.
- Use HTML5 attributes.
- Perform server-side validation.
- Give clear input instructions.

Conclusion:

Client-side validation improves efficiency and user experience but must always be combined with server-side validation.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 3: CSS Box Model and Layout Breaking

The CSS Box Model describes how every HTML element occupies space.

Components:

- Content
- Padding
- Border
- Margin

Positioning:

- Static
- Relative
- Absolute
- Fixed
- Sticky

Causes of layout breaking:

- Fixed widths.
- Large images.
- Incorrect positioning.
- Missing media queries.
- Overflowing content.

Media Query:

```
@media(max-width:768px){  
  .container{display:block;width:100%;}  
}
```

Best practices:

- Use Flexbox/Grid.
- Use relative units.
- Apply box-sizing:border-box.
- Responsive images.
- Test across devices.

Conclusion:

Understanding the box model and media queries helps create stable, responsive layouts.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for

updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Question 4: Document Object Model (DOM)

The Document Object Model (DOM) represents an HTML document as a tree of objects that JavaScript can access and modify dynamically.

Role:

- Change content.
- Modify styles.
- Add/remove elements.
- Handle events.

Example:

HTML:

```
<p id="message">Welcome Student!</p>
<button onclick="changeText()">Click</button>
```

JavaScript:

```
function changeText(){
  document.getElementById("message").innerHTML="Welcome to Web Programming";
}
```

Improvements:

- Update only required elements.
- Minimize DOM manipulation.
- Use event delegation.
- Show loading indicators.
- Validate user input.

Conclusion:

DOM manipulation enables interactive webpages without reloading the entire page.